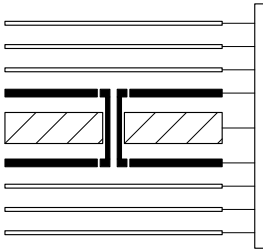
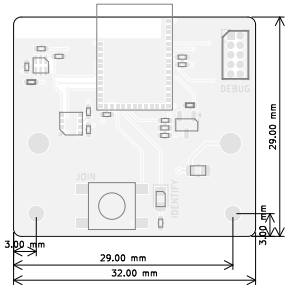


Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen			Legend	GBR
	F.Mask	0.01mm		Solder Mask	GBR
Copper	**F.Cu**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Core		1.51mm	FR4	Dielectric	
Copper	**B.Cu**	**0.035mm (1.00oz)**		**Signal**	**GBR**
	B.Mask	0.01mm		Solder Mask	GBR
	B.Silkscreen			Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.6mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [%]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLUE.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 150 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY NULL.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

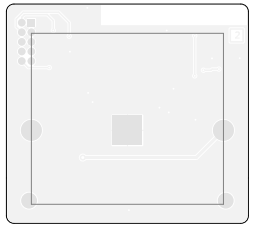
BOARD SIZE	32.000 × 29.000 mm
BOARD THICKNESS	1.600 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.125 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	2.100 mm
ANNULAR RING	0.100 mm
COPPER TO HOLE	0.125 mm
COPPER TO EDGE	0.125 mm
HOLE TO HOLE	0.250 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

12) CONFIRM SPACE WIDTHS AND SPACINGS.

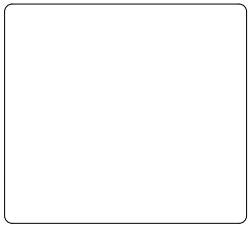
All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Kampis-Elektroecke		Variant: PRELIMINARY	Git Hash: 753f6cf
	Board Name: BeeLight		Project Name: BeeLight	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: BeeLight.kicad_pcb	Designer: Daniel Kampert	Date: 2025-05-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 1 of 7



All dimensions are in millimeters unless otherwise specified.

Comments:	Company: **Kampis-Elektroecke**		Variant: PRELIMINARY	Git Hash: 753f6cf
	Board Name: **BeeLight**		Project Name: **BeeLight**	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: BeeLight.kicad_pcb	Designer: Daniel Kampert	Date: 2025-05-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **2** of **7**



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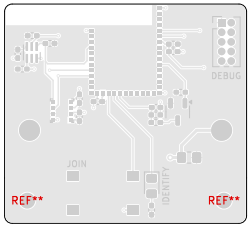
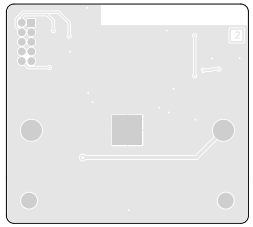
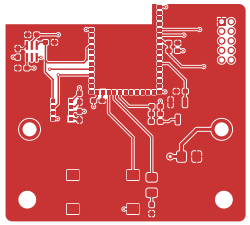
	Board Name: **BeeLight**		Project Name: **BeeLight**	
Sheet Title: Drill Drawing	File Name: BeeLight.kicad_pcb	Designer: Daniel Kampert	Date: 2025-05-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **3** of **7**



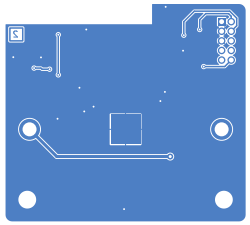
Comments:	Company: **Kampis-Elektroecke**		Variant: PRELIMINARY	Git Hash: 753f6cf
	Board Name: **BeeLight**		Project Name: **BeeLight**	
Sheet Title: Top Test Points (Scale 1:1)	File Name: BeeLight.kicad_pcb	Designer: Daniel Kampert	Date: 2025-05-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **4** of **7**



Comments:	Company: **Kampis-Elektroecke**		Variant: PRELIMINARY	Git Hash: 753f6cf
	Board Name: **BeeLight**		Project Name: **BeeLight**	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: BeeLight.kicad_pcb	Designer: Daniel Kampert	Date: 2025-05-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **5** of **7**



Comments:	Company: **Kampis-Elektroecke**		Variant: PRELIMINARY	Git Hash: 753f6cf
	Board Name: **BeeLight**		Project Name: **BeeLight**	
Sheet Title: F.Cu (Scale 1:1)	File Name: BeeLight.kicad_pcb	Designer: Daniel Kampert	Date: 2025-05-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **6** of **7**



Comments:	Company: **Kampis-Elektroecke**		Variant: PRELIMINARY	Git Hash: 753f6cf
	Board Name: **BeeLight**		Project Name: **BeeLight**	
Sheet Title: B.Cu (Scale 1:1)	File Name: BeeLight.kicad_pcb	Designer: Daniel Kampert	Date: 2025-05-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **7** of **7**